

Indian Analytics Job Market Intelligence

Capstone Project — Data Science PGC

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1. Business Problem

Analytics hiring in India is hard to read from the outside. Job titles are inconsistent, most postings never mention a salary, and there's no easy way to tell which skills actually get paid more versus which ones just show up a lot because everyone lists them.

For this project I collected live job postings from the Indian job market and tried to answer three things: what does the market actually pay, which skills carry a real salary premium, and does the market split into distinct types of analytics roles.

2. Data Collection

I used the Adzuna Jobs API for the Indian market (country code 'in'). I picked an API over scraping HTML directly because Adzuna publishes an official API for this data, and scraping the same site's pages instead would go against their terms.

Collection ran in two passes. A national pass covered 14 search terms (data analyst, business analyst, data scientist, etc.) going 10 pages deep. A second pass queried 10 major cities directly with 3 core search terms, because a single national search can't be paged deep enough to get past 1,000 unique postings on its own.

- Raw rows collected: 9,905
- Duplicates removed (same job under different search terms): 2,938
- Unique postings saved: 6,967
- Fields per posting: title, company, location, description, salary range, contract type, posting date, and a redirect URL

3. Data Cleaning

3.1 Locations

The city field had 96 different values, and a lot of that was the same place written differently — Bangalore/Bengaluru, or Gurgaon/Noida/Ghaziabad which are all really Delhi NCR. I built a keyword-based mapping to group these into proper markets, checking title, description, and region text together so a posting with a missing city could still be placed from other fields.

For the Tier 1 / Tier 2 split I used the Government of India's HRA city classification rather than guessing. Under that standard, Ahmedabad and Surat are Tier 1, not Tier 2, so I moved them up from my first attempt.

- Tier 1 cities: 5,222 postings (75%)
- Tier 2 cities: 176 postings
- Unknown (no location data at all): 1,569 postings (22.5%)

The 1,569 unknown postings still have complete titles and descriptions, so they're used everywhere except location-based comparisons.

3.2 Salary

1,363 of the 6,967 postings (about 20%) list a salary. That's a genuine finding on its own — most Indian job postings simply don't disclose pay. One thing that surprised me: every single salary figure in this dataset is marked as employer-stated, not a platform estimate. Adzuna sometimes fills in an estimated salary when a posting doesn't give one, but that didn't happen anywhere in this pull, so the 1,363 numbers I do have are trustworthy.

I took the midpoint of each posting's salary_min and salary_max as a single comparable figure, and capped the top and bottom 1% to stop a handful of extreme values (data entry errors, monthly figures typed into annual fields) from skewing every average.

3.3 Skills

There's no 'skills' column in the raw data — I built one by searching each posting's title and description for keywords. I started from what I already know analyst roles need (SQL, Excel, Power BI, Tableau, Python, statistics), then checked every candidate against the actual dataset before deciding what to keep.

This caught two mistakes. I'd assumed 'R' would be a common skill; it appeared in only 6 postings out of 6,967, so I dropped it. And a term I was about to add, 'Scala', showed 777 hits — until I checked and found it was matching inside the word 'scalable', not the programming language. I dropped that too.

Final list: SQL, Python, Excel, Power BI, Tableau, Cloud (AWS/Azure/GCP/Snowflake/Databricks), Big Data (Spark/Hadoop/Hive/Kafka), Machine Learning, ETL, Statistics, and Visualisation.

Important limitation: Adzuna's description field is a truncated excerpt, not the full job ad. 3,663 of 6,967 postings (53%) don't mention any of these 11 skills at all — not because the job doesn't need them, but because the text simply got cut off before that part. So every skill count in this report is a floor, not an exact figure.

3.4 Seniority and role family

Both were built from the job title using keyword rules, checked in order from most specific to least (so 'Business Intelligence Analyst' gets caught as BI/Reporting before the generic 'analyst' rule can grab it).

- Seniority: Mid 4,206 · Senior 1,251 · Lead/Manager 1,063 · Entry 264 · Leadership 183
- Role family: Data Analysis 1,890 · Data Engineering 1,069 · Business Analysis 767 · BI/Reporting 646 · Data Science/ML 429 · Other 2,394

The 'Other' bucket is large because some of my search terms pulled in non-analytics jobs — Relationship Manager, Sales Manager, Assistant Professor — that Adzuna still matched. I left these as Other rather than forcing them into a family they don't belong in.

4. Key Finding: Skill Demand vs. Skill Premium

This is the main result of the project. For each skill, I compared the median salary of postings that mention it against postings that don't. The pattern is consistent: the most commonly listed skills are not the ones that pay more.

Skill	Postings	Median salary (with)	Median salary (without)	Premium
Machine Learning	262	₹16,50,000	₹9,00,000	+83.3%
ETL	741	₹13,00,000	₹8,50,000	+52.9%
Cloud	1,239	₹12,75,000	₹8,50,000	+50.0%
Big Data	333	₹13,00,000	₹9,00,000	+44.4%
Statistics	232	₹12,50,000	₹9,00,000	+38.9%
Python	638	₹12,50,000	₹9,00,000	+38.9%
SQL	1,088	₹11,00,000	₹8,50,000	+29.4%
Power BI	452	₹11,00,000	₹9,00,000	+22.2%
Tableau	200	₹10,00,000	₹9,50,000	+5.3%
Visualisation	714	₹9,06,400	₹9,50,000	-4.6%
Excel	554	₹7,00,000	₹10,00,000	-30.0%

Machine Learning is mentioned in only 3.8% of postings but carries the largest premium in the dataset — 83.3% higher than postings without it. Excel, on the other hand, is mentioned often but is associated with a 30% lower median salary than postings that don't mention it. This doesn't mean Excel lowers pay — it means Excel gets mentioned in lower-paid role types (MIS, basic reporting), so the skill is a marker of the role rather than a driver of pay by itself.

A correlation heatmap backed this up from a completely different angle: Machine Learning had the strongest positive correlation with salary (0.14) of any skill, and it was also the highest premium in the table above. Two different methods, same answer.

5. Geography and Seniority

Bengaluru has both the most postings (1,167) and the highest median salary among Tier 1 cities. One result that needed a second look: Pune's median salary (₹10,00,000) came out higher than Mumbai's (₹8,00,000), which doesn't match cost of living. Checking the role mix explained it — 42% of Pune's salaried postings mention data engineering or cloud skills, against 22% in Mumbai. Pune's higher pay is a role-mix effect, not a city effect.

Seniority moves mostly as expected — Entry lowest, rising through Mid and Senior — except Lead/Manager dips slightly below Senior. Checking the actual titles in that group showed why: some non-analytics titles (Relationship Manager, Assistant Manager, Sales Manager) landed in Lead/Manager because their titles contain the word 'manager', which pulls the group's median down. This is a labelling limitation, not a real market pattern.

6. Model: Role Archetype Clustering

I used k-means clustering to group postings by their skill profile, restricted to the 3,304 postings that mention at least one skill (grouping the other 3,663 zero-skill postings would just create one meaningless empty cluster). I tested cluster counts from 2 to 8 using silhouette score and picked k=8, the best-scoring option in that range.

Segment	Postings	Median salary	Defining skills
Advanced ML Engineering	197	₹15,00,000	ML, Cloud, Python
Cloud Data Pipeline	813	₹14,50,000	Cloud, ETL, Python
Statistics & ML Analytics	208	₹12,75,000	Statistics, ML, Python
Big Data Engineering	300	₹13,00,000	Big Data, Cloud, SQL
Cloud & SQL Engineering	623	₹10,90,000	SQL, Cloud, ETL
Power BI Reporting	534	₹10,00,000	Visualisation, Power BI, SQL
Tableau Reporting	200	₹10,00,000	Tableau, Visualisation, SQL
Standard MIS / Excel Reporting	429	₹6,50,000	Excel, SQL, Visualisation
No skill listed in excerpt	3,663	—	—

The segments separate cleanly by pay, from ₹6,50,000 up to ₹15,00,000 — more than double. A few segments overlap in skill profile (Cloud & SQL Engineering and Big Data Engineering are both engineering-flavoured),

which suggests the true number of distinct role types in this market may be smaller than 8, but grouping this finely still keeps every segment interpretable.

I chose not to build a salary regression model. With only 1,363 salaried postings and skill flags that undercount due to truncated text, a regression model tested at close to zero cross-validated R^2 — meaning it wasn't predicting anything better than guessing the average. Reporting a model that doesn't work isn't more thorough than leaving it out, so given the time available I focused on the clustering model, which is genuinely useful.

7. Dashboard (Power BI)

Built on the cleaned dataset with two pages: a Market Overview page (KPI cards, postings-by-segment chart, a Decomposition Tree breaking salary down by role family, seniority and city) and a Segment Detail page (a filterable table of individual postings).

- KPI cards: Total Postings, Median Salary, Salaried Postings, Skill Coverage %, Avg Skills per Posting
- Decomposition Tree — an AI visual letting a viewer click through role family → seniority → city and see the median salary update at each level
- Sync slicer — a city filter that stays in sync across both pages
- Bookmarks — 'Bengaluru View' and 'All Markets' buttons that jump between filter states
- Row-Level Security — a Bengaluru_Recruiter role that restricts the whole report to Bengaluru postings only (see screenshot)

Two things I attempted but didn't finish: the Key Influencers AI visual didn't return usable results, most likely because the salary field is 80% blank and the visual struggles to find a pattern in that little data. Drill-through from the segment chart to the detail page also wasn't completed in the time available.

8. Recommendations

- For candidates: prioritise skills that combine low-to-moderate demand with a real premium — Machine Learning, ETL, Cloud, and Big Data all fit this. Skills like Excel and general visualisation are common but don't carry a premium on their own; they're table stakes, not differentiators.
- For employers: postings that combine multiple engineering skills (cloud + ETL + Python) cluster at the top of the pay scale and represent the segment competing hardest for talent. Reporting-focused roles (Excel/Power BI/Tableau) sit at the bottom and may be easier to fill.
- For the market generally: only 1 in 5 postings discloses a salary. That's a transparency gap on its own — job seekers in this market are mostly negotiating blind.

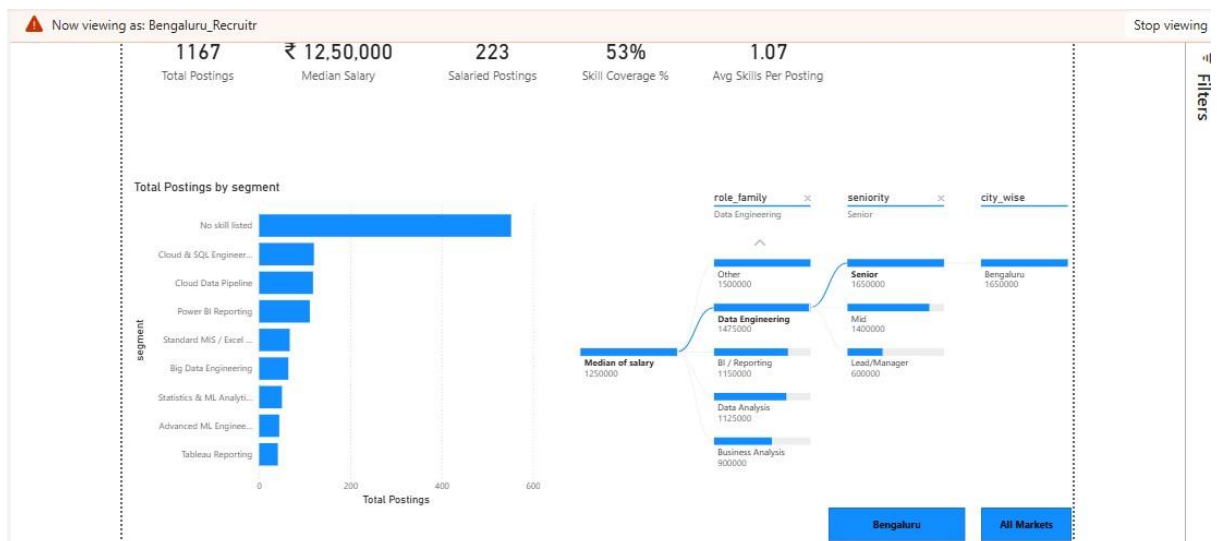
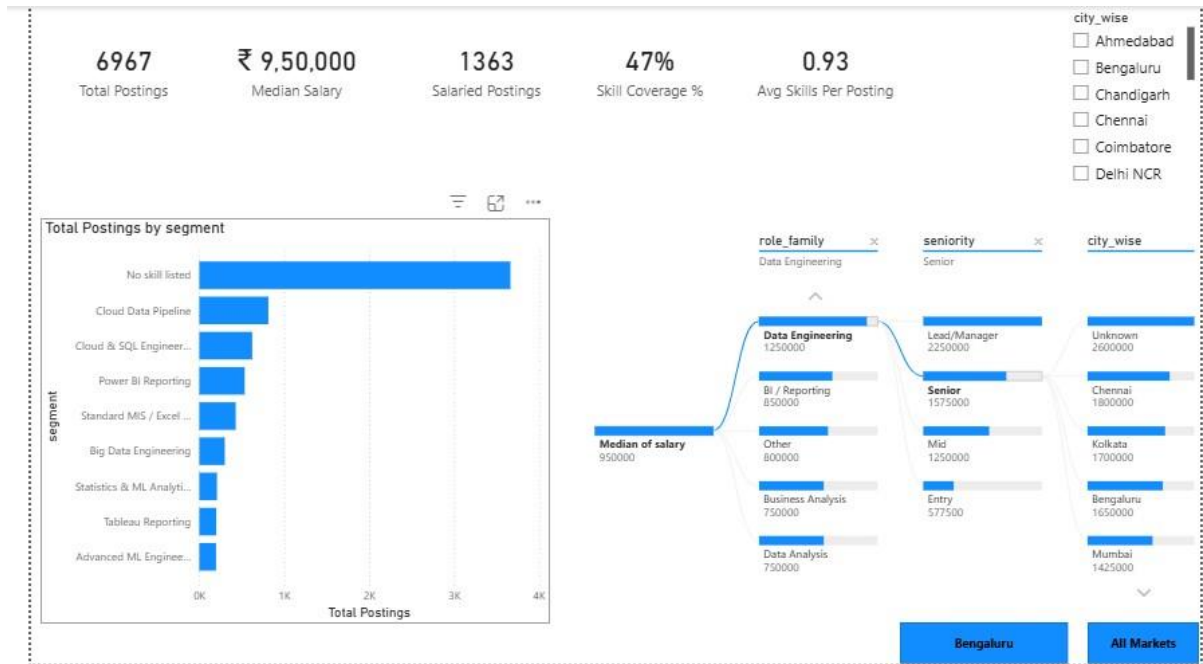
9. Limitations

- Only 20% of postings have a salary figure — every salary-based comparison in this report runs on that smaller sample, not the full 6,967.
- 53% of postings show zero detected skills because Adzuna's description field is a truncated excerpt. Skill counts and premiums here are a floor, not a ceiling.
- 22.5% of postings have no recorded location and are excluded from city-level comparisons.
- Search terms pulled in some non-analytics job titles (Sales Manager, Relationship Manager) that landed in 'Other' or affected the Lead/Manager seniority group.
- The Key Influencers visual and the drill-through dashboard feature were attempted but not completed given the project timeline.

Appendix: Files Submitted

- 01_data_extraction.py — Adzuna API extraction script
- 02_cleaning_eda_modelling.ipynb — cleaning, EDA, and clustering model
- cleaned_job_postings.csv — the clean dataset
- powerbi_dataset_job_posting.csv — dataset used in the dashboard
- job_market_dashboard.pbix — the Power BI file

- Video walkthrough:
- <https://drive.google.com/file/d/1gx2-CXmiZAZ3X-5NSUxVXt11NB1HusHg/view?usp=sharing>



< Back to report

title	company	city_wise	seniority	Sum of salary	segment
"SQL Developer"	OJCommerce	Chennai	Mid	600000	Cloud & SQL Enginee
.net Developer	Technogen India PvtLtd	Hyderabad	Mid	4250000	Cloud & SQL Enginee
.net developer ms-sql developer	MYLAS IT PRIVATE LIMITED	Unknown	Mid	150000	Cloud & SQL Enginee
.NET MVC Developer - Asp/mvc/sql Server (Team Leader)	Vexil Infotech	Lucknow	Lead/Manager	300000	Cloud & SQL Enginee
Accounts Executive	Hunarstreet Technologies Pvt Ltd	Mumbai	Mid	400000	No skill listed
Admissions Head	Regul Solutions	Unknown	Leadership	1750000	No skill listed
Adobe Enterprise Manager	Ele1 Consultant	Kolkata	Lead/Manager	1250000	No skill listed
Advanced SQLSnowflake_5Years	Zorba Consulting India	Bengaluru	Mid	1500000	Cloud & SQL Enginee
AEP/CJA Developer	Weekday	Unknown	Mid	2500000	No skill listed
AI & Process Transformation Specialist	JB Group	Delhi NCR	Mid	1200000	No skill listed
AI / ML Automation Engineer	Corvi Technologies	Delhi NCR	Mid	2250000	No skill listed
AI Automation Engineer	E2M Solutions Pvt. Ltd.	Ahmedabad	Mid	1000000	Advanced ML Enginee
AI Developer L4	Praveza Consultancy Private Limited	Mumbai	Mid	1650000	No skill listed
AI Engineer	DS Group	Unknown	Mid	906400	Standard MIS / Excel I
AI Engineer - R&D	Mihup	Kolkata	Mid	4500000	No skill listed
AI Product Engineer	Flipr	Bengaluru	Mid	1650000	Cloud Data Pipeline
AI QA	E2M Solutions Pvt. Ltd.	Ahmedabad	Mid	550000	No skill listed
AI/ML	Acme Services Private Limited	Ahmedabad	Mid	2300000	No skill listed
AI/ML Engineer	NirwanaAI	Delhi NCR	Mid	3100000	Advanced ML Enginee
AI/ML Engineer	r3 Consultant	Delhi NCR	Mid	1250000	Advanced ML Enginee
Total				1591669231	